

BOMI COMMUNITY COLLEGE

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TOPIC:

Maternal Fetal Distress

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Obstetric and Gynecology in Nursing

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Introduction

Maternal fetal distress is a serious condition that occurs when the fetus does not receive enough oxygen before or during labor. It is considered an obstetric emergency because prolonged lack of oxygen can harm the baby and may also affect the health of the mother. Early recognition and prompt management are important to prevent complications and improve the outcome for both mother and child. Fetal distress is often identified through abnormal fetal heart rate patterns, reduced fetal movements, or meconium-stained amniotic fluid.

Definition

Maternal Distress

Maternal distress refers to any physical or emotional condition in the mother during pregnancy or labor that negatively affects her health and may reduce the supply of oxygen and nutrients to the fetus.

Fetal Distress

Fetal distress is a condition in which the fetus shows signs of inadequate oxygen supply (hypoxia) before or during labor. It indicates that the fetus is not coping well in the uterine environment and requires immediate attention.

General Considerations

The following points are important in understanding maternal fetal distress:

It is an emergency condition that requires immediate assessment and intervention.

Early detection helps prevent serious complications and fetal death.

Maternal and fetal conditions should always be assessed together because problems affecting the mother can directly affect the fetus.

Continuous monitoring of fetal heart rate and maternal vital signs is essential.

Prompt treatment can improve outcomes for both mother and baby.

In severe cases, immediate delivery may be necessary through assisted vaginal delivery or cesarean section.

Etiology (Causes)

Maternal fetal distress may result from maternal, placental, umbilical cord, or fetal factors.

1. Maternal Causes

These conditions reduce the amount of oxygen reaching the fetus.

a. Hypertension (High Blood Pressure)

High blood pressure reduces blood flow to the placenta, thereby decreasing oxygen delivery to the fetus.

b. Severe Anemia

Anemia reduces the oxygen-carrying capacity of the mother's blood and limits oxygen supply to the fetus.

c. Diabetes Mellitus

Diabetes may affect placental function and lead to fetal compromise.

d. Maternal Infection

Maternal fever and infection increase fetal stress and oxygen demand.

e. Hypotension (Low Blood Pressure)

Low blood pressure decreases placental blood flow and fetal oxygenation.

f. Respiratory Diseases

Conditions such as asthma and pneumonia can reduce maternal oxygen levels.

g. Excessive Uterine Contractions

Overstimulation of the uterus, especially from excessive oxytocin use, can decrease blood flow to the fetus.

2. Placental Causes

a. Placental Abruption

Premature separation of the placenta from the uterine wall reduces oxygen and nutrient supply to the fetus.

b. Placenta Previa

Abnormal placement of the placenta may cause bleeding and compromise fetal oxygenation.

c. Placental Insufficiency

The placenta fails to provide adequate oxygen and nutrients to the fetus.

3. Umbilical Cord Causes

a. Cord Prolapse

The umbilical cord descends before the baby and becomes compressed.

b. Cord Compression

Pressure on the cord reduces blood flow between the mother and fetus.

c. Nuchal Cord

The umbilical cord wraps around the fetal neck, affecting circulation.

4. Fetal Causes

a. Prematurity

Premature babies often have limited ability to tolerate labor stress.

b. Congenital Abnormalities

Certain birth defects may interfere with oxygenation and fetal adaptation.

c. Intrauterine Growth Restriction (IUGR)

Poor fetal growth caused by inadequate oxygen and nutrient supply.

Clinical Features and Assessment

Clinical Features

The following signs may indicate maternal fetal distress.

1. Abnormal Fetal Heart Rate

The normal fetal heart rate ranges from 110–160 beats per minute.

Abnormal findings include:

Fetal tachycardia (above 160 beats/minute)

Fetal bradycardia (below 110 beats/minute)

Irregular heart rhythm

2. Reduced or Absent Fetal Movement

The mother may notice that the baby is moving less frequently than usual.

3. Meconium-Stained Amniotic Fluid

Green or brown-colored amniotic fluid may indicate fetal stress and oxygen deprivation.

4. Abnormal Uterine Contractions

Contractions may become excessively strong, prolonged, or frequent.

5. Maternal Symptoms

Depending on the cause, the mother may experience:

Vaginal bleeding

Severe abdominal pain

Fever

Difficulty breathing

High blood pressure

Assessment of Maternal Fetal Distress

1. History Taking

The healthcare provider should assess:

Maternal medical conditions

Pattern of fetal movement

Duration and progress of labor

Medication or drug use during labor

Vaginal bleeding or leaking amniotic fluid

2. Physical Examination

Maternal Assessment

Check:

Blood pressure

Pulse rate

Temperature

Respiratory rate

Oxygen saturation

Fetal Assessment

Check:

Fetal heart rate

Fetal movements

Uterine contractions

3. Electronic Fetal Monitoring

Continuous monitoring helps identify abnormal fetal heart rate patterns such as:

Late decelerations

Variable decelerations

Prolonged bradycardia

4. Ultrasound Examination

Ultrasound is used to assess:

Placental condition

Fetal growth

Amniotic fluid volume

5. Biophysical Profile (BPP)

The Biophysical Profile evaluates:

Fetal breathing movements

Fetal body movements

Fetal muscle tone

Amniotic fluid volume

Fetal heart rate response

6. Fetal Scalp Blood Sampling

This procedure may be performed during labor to assess fetal blood pH and determine the degree of oxygenation.

Conclusion

Maternal fetal distress is a serious obstetric condition resulting from inadequate oxygen supply to the fetus before or during labor. Early identification through careful assessment of maternal and fetal conditions is essential for preventing complications. Continuous monitoring, prompt intervention, and appropriate obstetric management can significantly improve outcomes for both the mother and the baby. Therefore, healthcare professionals must recognize the signs early and provide timely treatment to ensure safe delivery and healthy maternal and fetal outcomes.

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